

● EASY

● ALGEBRA

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# Linear functions

# 03

10 questions · ~15 min

Q1

ALGEBRA

For the linear function  $f$ , the graph of  $y = f(x)$  in the  $xy$ -plane has a slope of 7 and passes through the point  $(0, 5)$ . Which equation defines  $f$ ?

- A.  $f(x) = 5x$
- B.  $f(x) = 35x$
- C.  $f(x) = 7x + 5$
- D.  $f(x) = 12x + 5$

Q2

ALGEBRA

Marisol drove 3 hours from City A to City B. The equation below estimates the distance  $d$ , in miles, Marisol traveled after driving for  $t$  hours.

$$d = 45t$$

Which of the following does 45 represent in the equation?

- A. Marisol took 45 trips from City A to City B.
- B. The distance between City A and City B is 45 miles.
- C. Marisol drove at an average speed of about 45 miles per hour.
- D. It took Marisol 45 hours to drive from City A to City B.

**Q3**

ALGEBRA

The function  $f$  is defined by  $fx = 92x + 3$ . For what value of  $x$  does  $fx = 63$ ?

- A. 2
- B. 5
- C. 7
- D. 30

**Q4**

ALGEBRA

The function  $f$  is defined by  $f(x) = 4x - 3$ . What is the value of  $f(10)$ ?

- A. -30
- B. 37
- C. 40
- D. 43

**Q5****ALGEBRA**

$$d = 16 - \frac{x}{30}$$

The equation shown gives the estimated amount of diesel  $d$ , in gallons, that remains in the gas tank of a truck after being driven  $x$  miles, where  $0 \leq x \leq 480$ . What is the estimated amount of diesel, in gallons, that remains in the gas tank of the truck when  $x = 300$ ?

- A. 0
- B. 6
- C. 14
- D. 16

**Q6****ALGEBRA**

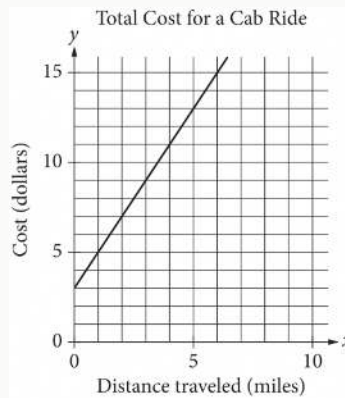
The function  $f$  is defined by  $f(x) = 25x + 30$ . What is the value of  $f(x)$  when  $x = 2$ ?

- A. 50
- B. 57
- C. 80
- D. 110

Josie purchased a pass that cost \$ 19.50 to tour a nature preserve. She also purchased 10 identical stickers to give to her nieces and nephews. The pass and the stickers cost \$ 62.00 total. What was the cost of one sticker?

- A. \$ 1.95
- B. \$ 4.25
- C. \$ 6.20
- D. \$ 8.15

The line graphed in the  $xy$ -plane below models the total cost, in dollars, for a cab ride,  $y$ , in a certain city during nonpeak hours based on the number of miles traveled,  $x$ .



According to the graph, what is the cost for each additional mile traveled, in dollars, of a cab ride?

- A. \$2.00
- B. \$2.60
- C. \$3.00
- D. \$5.00

Q9

GRID-IN ALGEBRA

The function  $f$  is defined by the equation  $fx = 7x + 2$ . What is the value of  $fx$  when  $x = 4$ ?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The function  $f$  is defined by  $fx = 80 - 6x$ . What is the value of  $f7$ ?

A. 13

B. 38

C. 74

D. 81